



SAFE LANDING ZONE DETECTION FOR UAVS EMERGENCY LANDING USING SEMANTIC SEGMENTATION

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MOTIVATION

- Emergency landing is critical when UAVs face power loss, fault, or mission failure.
- Manual landing decisions are slow and risky in cluttered environments.
- GPS-only mapping does not capture local terrain hazards.
- A fast vision-based landing analysis system can improve autonomy and safety.

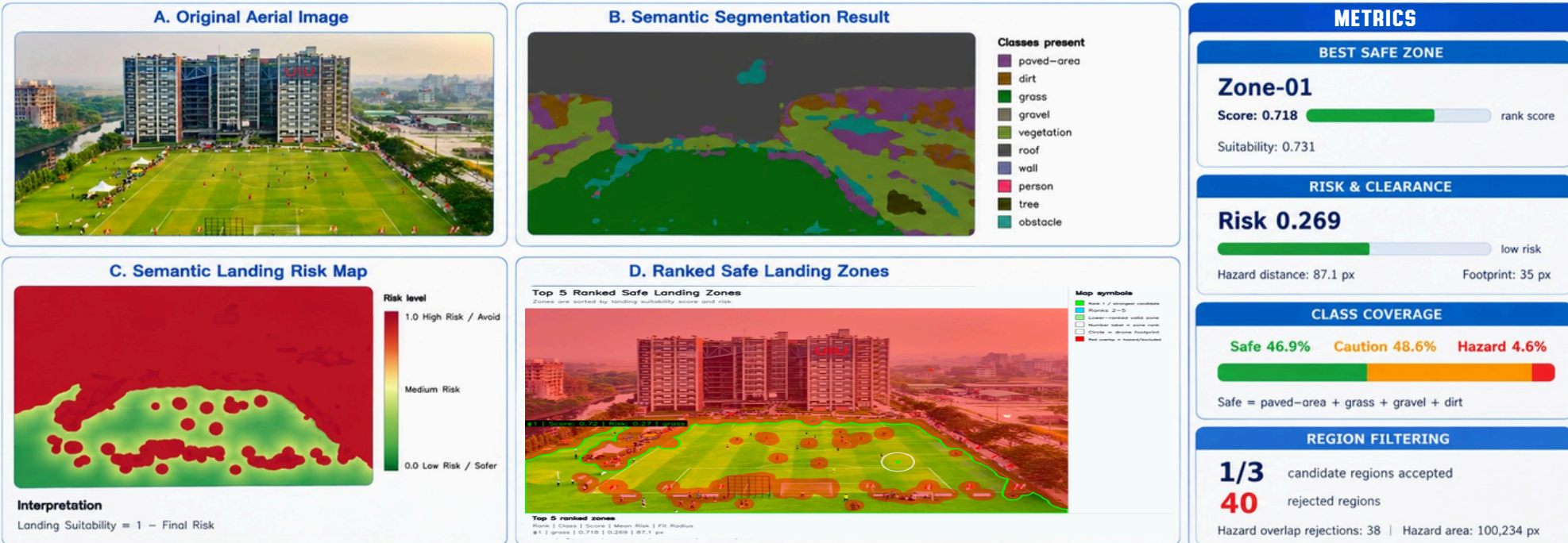
PROBLEM & OBJECTIVES

- Detect terrain semantics from aerial RGB imagery.
- Separate safe, caution, and hazardous regions.
- Estimate candidate landing zones using area and clearance.
- Rank the best landing sites for real-time decision support.

PROPOSED METHODOLOGY / SYSTEM ARCHITECTURE



RESULT ANALYSIS



KEY CONTRIBUTIONS

- RGB-image-based autonomous landing analysis
- Interpretable segmentation-driven risk assessment
- Ranked safe-zone selection for emergency landing
- Lightweight workflow suitable for future UAV integration



CONCLUSION & FUTURE WORK

- The current prototype can identify potential landing regions from a single aerial image.
- Semantic understanding improves safety compared with naive open-space detection.
- Future work: add depth / elevation, ROS2-Gazebo landing simulation, onboard deployment, and multi-modal risk estimation.



Reference

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